

5.3 Benchmarking Comparison Component

typescript

TypeScript

```
// components/review/BenchmarkingComparison.tsx
import React, { useState } from 'react';
import { Card, CardHeader, CardTitle, CardContent } from
'@/components/ui/card';
import { Badge } from '@components/ui/badge';
import { Button } from '@components/ui/button';
import { Tabs, TabsContent, TabsList, TabsTrigger } from
'@/components/ui/tabs';
import {
  TrendingUp,
  TrendingDown,
  Minus,
  Trophy,
  Target,
  BarChart3,
  LineChart,
  Award,
  Users,
  Building2,
  Globe2
} from 'lucide-react';
import {
  LineChart as RechartsLine,
  Line,
  BarChart,
  Bar,
  XAxis,
  YAxis,
  CartesianGrid,
  Tooltip,
```

```
    Legend,  
    ResponsiveContainer,  
    RadarChart,  
    PolarGrid,  
    PolarAngleAxis,  
    PolarRadiusAxis,  
    Radar  
} from 'recharts';  
  
interface BenchmarkMetric {  
    metric_name: string;  
    current_value: number;  
    peer_average: number;  
    industry_average: number;  
    best_in_class: number;  
    unit: string;  
    trend: 'improving' | 'declining' | 'stable';  
    percentile_rank: number;  
}  
  
interface PeerComparison {  
    venue_name: string;  
    venue_type: string;  
    capacity: number;  
    region: string;  
    metrics: {  
        [key: string]: number;  
    };  
}  
  
interface HistoricalTrend {  
    period: string;  
    current_venue: number;  
    peer_average: number;  
    industry_average: number;  
}
```

```
export const BenchmarkingComparison: React.FC<{
  eventId: string;
  venueId: string;
}> = ({ eventId, venueId }) => {
  const [selectedCategory, setSelectedCategory] =
    useState<string>('operational');
  const [comparisonScope, setComparisonScope] = useState<'peer' |
    'industry' | 'global'>('peer');

  // Mock data - replace with API calls
  const benchmarkMetrics: BenchmarkMetric[] = [
    {
      metric_name: 'Incident Response Time',
      current_value: 4.2,
      peer_average: 12.5,
      industry_average: 15.3,
      best_in_class: 3.1,
      unit: 'minutes',
      trend: 'improving',
      percentile_rank: 92
    },
    {
      metric_name: 'Incident Prevention Rate',
      current_value: 67,
      peer_average: 45,
      industry_average: 38,
      best_in_class: 72,
      unit: '%',
      trend: 'improving',
      percentile_rank: 88
    },
    {
      metric_name: 'Guest Satisfaction Score',
      current_value: 4.7,
      peer_average: 4.2,
```

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        industry_average: 3.9,
        best_in_class: 4.8,
        unit: '/5',
        trend: 'stable',
        percentile_rank: 85
    },
    {
        metric_name: 'Equipment Downtime',
        current_value: 2.1,
        peer_average: 8.7,
        industry_average: 12.3,
        best_in_class: 1.5,
        unit: 'hours/month',
        trend: 'improving',
        percentile_rank: 94
    },
    {
        metric_name: 'Staff Efficiency',
        current_value: 156,
        peer_average: 112,
        industry_average: 98,
        best_in_class: 168,
        unit: 'tasks/hour',
        trend: 'improving',
        percentile_rank: 79
    }
];

const peerComparisons: PeerComparison[] = [
    {
        venue_name: 'Our Venue',
        venue_type: 'NFL Stadium',
        capacity: 70000,
        region: 'Southeast',
        metrics: {
            response_time: 4.2,

```

```
        prevention_rate: 67,  
        satisfaction: 4.7,  
        downtime: 2.1,  
        efficiency: 156  
    },  
    },  
    {  
        venue_name: 'Peer Stadium A',  
        venue_type: 'NFL Stadium',  
        capacity: 72000,  
        region: 'Southeast',  
        metrics: {  
            response_time: 11.8,  
            prevention_rate: 48,  
            satisfaction: 4.3,  
            downtime: 7.2,  
            efficiency: 118  
        }  
    },  
    {  
        venue_name: 'Peer Stadium B',  
        venue_type: 'NFL Stadium',  
        capacity: 68000,  
        region: 'Southeast',  
        metrics: {  
            response_time: 13.2,  
            prevention_rate: 42,  
            satisfaction: 4.1,  
            downtime: 9.1,  
            efficiency: 105  
        }  
    },  
    {  
        venue_name: 'Best in Class',  
        venue_type: 'NFL Stadium',  
        capacity: 75000,
```

```
    region: 'West',
    metrics: {
      response_time: 3.1,
      prevention_rate: 72,
      satisfaction: 4.8,
      downtime: 1.5,
      efficiency: 168
    }
  }
];
```

```
const historicalTrends: HistoricalTrend[] = [
  { period: 'Q1 2024', current_venue: 15.2, peer_average: 14.8,
    industry_average: 16.1 },
  { period: 'Q2 2024', current_venue: 12.1, peer_average: 13.9,
    industry_average: 15.7 },
  { period: 'Q3 2024', current_venue: 8.7, peer_average: 13.2,
    industry_average: 15.4 },
  { period: 'Q4 2024', current_venue: 5.3, peer_average: 12.8,
    industry_average: 15.2 },
  { period: 'Q1 2025', current_venue: 4.2, peer_average: 12.5,
    industry_average: 15.3 }
];
```

```
const radarData = [
  {
    metric: 'Response\nTime',
    current: 95,
    peer: 72,
    industry: 65,
    fullMark: 100
  },
  {
    metric: 'Prevention\nRate',
    current: 88,
    peer: 65,
```

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        industry: 58,
        fullMark: 100
    },
    {
        metric: 'Guest\nSatisfaction',
        current: 94,
        peer: 84,
        industry: 78,
        fullMark: 100
    },
    {
        metric: 'Uptime',
        current: 92,
        peer: 78,
        industry: 70,
        fullMark: 100
    },
    {
        metric: 'Efficiency',
        current: 85,
        peer: 68,
        industry: 62,
        fullMark: 100
    }
];

const getTrendIcon = (trend: string) => {
    switch (trend) {
        case 'improving':
            return <TrendingUp className="h-4 w-4 text-[#9AAF45]" />;
        case 'declining':
            return <TrendingDown className="h-4 w-4 text-red-500" />;
        default:
            return <Minus className="h-4 w-4 text-[#5A6A72]" />;
    }
};

```

```
const getPerformanceColor = (percentile: number) => {
  if (percentile >= 90) return 'text-[#9AAF45] bg-[#9AAF45]/10 border-[#9AAF45]';
  if (percentile >= 75) return 'text-[#3A7B8A] bg-[#3A7B8A]/10 border-[#3A7B8A]';
  if (percentile >= 50) return 'text-orange-500 bg-orange-50 border-orange-500';
  return 'text-red-500 bg-red-50 border-red-500';
};

const getPerformanceBadge = (percentile: number) => {
  if (percentile >= 90) return { icon: <Trophy className="h-4 w-4" />, label: 'Elite' };
  if (percentile >= 75) return { icon: <Award className="h-4 w-4" />, label: 'Excellent' };
  if (percentile >= 50) return { icon: <Target className="h-4 w-4" />, label: 'Average' };
  return { icon: <TrendingDown className="h-4 w-4" />, label: 'Below Average' };
};

return (
  <div className="space-y-6">
    { /* Header Controls */ }
    <Card className="border-[#1B3A4B]">
      <CardContent className="p-4">
        <div className="flex items-center justify-between">
          <div className="flex items-center gap-4">
            <div className="flex items-center gap-2">
              <BarChart3 className="h-5 w-5 text-[#1B3A4B]" />
              <span className="font-semibold text-[#1B3A4B]">Comparison Scope:</span>
            </div>
            <div className="flex gap-2">
              <Button
```



```

        size="sm"
        variant={comparisonScope === 'peer' ? 'default'
: 'outline'}
        onClick={() => setComparisonScope('peer')}
        className={comparisonScope === 'peer' ?
'bg-[#3A7B8A]' : ''}
      >
        <Users className="h-4 w-4 mr-2" />
        Peer Venues
      </Button>
      <Button
        size="sm"
        variant={comparisonScope === 'industry' ?
'default' : 'outline'}
        onClick={() => setComparisonScope('industry')}
        className={comparisonScope === 'industry' ?
'bg-[#3A7B8A]' : ''}
      >
        <Building2 className="h-4 w-4 mr-2" />
        Industry
      </Button>
      <Button
        size="sm"
        variant={comparisonScope === 'global' ?
'default' : 'outline'}
        onClick={() => setComparisonScope('global')}
        className={comparisonScope === 'global' ?
'bg-[#3A7B8A]' : ''}
      >
        <Globe2 className="h-4 w-4 mr-2" />
        Global
      </Button>
    </div>
  </div>
</div>
</CardContent>

```

```

</Card>

{/* Performance Summary Cards */}
<div className="grid grid-cols-3 gap-6">
  <Card className="border-[#9AAF45]">
    <CardContent className="p-6">
      <div className="flex items-center justify-between
mb-2">
        <Trophy className="h-8 w-8 text-[#9AAF45]" />
        <span className="text-3xl font-bold
text-[#9AAF45]">Top 10%</span>
      </div>
      <p className="text-sm text-[#5A6A72]">Overall
Performance Ranking</p>
      <p className="text-xs text-[#5A6A72] mt-2">
        Across {benchmarkMetrics.length} key metrics
      </p>
    </CardContent>
  </Card>

  <Card className="border-[#3A7B8A]">
    <CardContent className="p-6">
      <div className="flex items-center justify-between
mb-2">
        <TrendingUp className="h-8 w-8 text-[#3A7B8A]" />
        <span className="text-3xl font-bold
text-[#3A7B8A]">+23%</span>
      </div>
      <p className="text-sm
text-[#5A6A72]">Quarter-over-Quarter Improvement</p>
      <p className="text-xs text-[#5A6A72] mt-2">
        Average across all categories
      </p>
    </CardContent>
  </Card>

```

```

<Card className="border-[#1B3A4B]">
  <CardContent className="p-6">
    <div className="flex items-center justify-between
mb-2">
      <Target className="h-8 w-8 text-[#1B3A4B]" />
      <span className="text-3xl font-bold
text-[#1B3A4B]">4/5</span>
    </div>
    <p className="text-sm text-[#5A6A72]">Metrics Above
Peer Average</p>
    <p className="text-xs text-[#5A6A72] mt-2">
      Leading in key operational areas
    </p>
  </CardContent>
</Card>
</div>

{/* Detailed Metrics Comparison */}
<Card className="border-[#1B3A4B]">
  <CardHeader className="bg-gradient-to-r from-[#1B3A4B]
to-[#2E5A6B]">
    <CardTitle className="text-white">Detailed Performance
Metrics</CardTitle>
  </CardHeader>
  <CardContent className="p-6">
    <div className="space-y-4">
      {benchmarkMetrics.map(metric => {
        const performance =
getPerformanceBadge(metric.percentile_rank);
        return (
          <div
            key={metric.metric_name}
            className="p-4 border-2 border-[#F4F6F7]
rounded-lg hover:border-[#3A7B8A] transition-colors"
          >

```

```

        <div className="flex items-start
justify-between mb-3">
            <div className="flex-1">
                <div className="flex items-center gap-3
mb-2">
                    <h4 className="font-semibold
text-[#1B3A4B]">{metric.metric_name}</h4>
                    {getTrendIcon(metric.trend)}
                    <Badge
                        variant="outline"

className={getPerformanceColor(metric.percentile_rank)}
                    >
                        {performance.icon}
                        <span
className="ml-1">{performance.label}</span>
                    </Badge>
                </div>
                <div className="text-xs text-[#5A6A72]">
                    Percentile Rank:
                    {metric.percentile_rank}th
                </div>
            </div>
            <div className="text-right">
                <div className="text-2xl font-bold
text-[#1B3A4B]">
                    {metric.current_value}
                    <span className="text-sm font-normal
text-[#5A6A72] ml-1">
                        {metric.unit}
                    </span>
                </div>
            </div>
        </div>

        {/* Comparison Bars */}

```

```

        <div className="space-y-2">
          <div>
            <div className="flex items-center
justify-between mb-1">
              <span className="text-xs
text-[#5A6A72]">Your Venue</span>
              <span className="text-xs font-semibold
text-[#1B3A4B]">
                {metric.current_value} {metric.unit}
              </span>
            </div>
            <div className="h-2 bg-[#F4F6F7]
rounded-full overflow-hidden">
              <div
                className="h-full bg-[#9AAF45]"
                style={{
                  width: `${(metric.current_value /
metric.best_in_class) * 100}%`
                }}
              />
            </div>
          </div>

          <div>
            <div className="flex items-center
justify-between mb-1">
              <span className="text-xs
text-[#5A6A72]">Peer Average</span>
              <span className="text-xs font-semibold
text-[#5A6A72]">
                {metric.peer_average} {metric.unit}
              </span>
            </div>
            <div className="h-2 bg-[#F4F6F7]
rounded-full overflow-hidden">
              <div

```

```

        className="h-full bg-[#3A7B8A]"
        style={{
            width: `${(metric.peer_average /
metric.best_in_class) * 100}%`
        }}
    />
</div>
</div>

<div>
    <div className="flex items-center
justify-between mb-1">
        <span className="text-xs
text-[#5A6A72]">Industry Average</span>
        <span className="text-xs font-semibold
text-[#5A6A72]">
            {metric.industry_average} {metric.unit}
        </span>
    </div>
    <div className="h-2 bg-[#F4F6F7]
rounded-full overflow-hidden">
        <div
            className="h-full bg-[#5A6A72]"
            style={{
                width: `${(metric.industry_average /
metric.best_in_class) * 100}%`
            }}
        />
    </div>
</div>

<div>
    <div className="flex items-center
justify-between mb-1">
        <span className="text-xs text-[#5A6A72]
flex items-center gap-1">

```

```

                <Trophy className="h-3 w-3" />
                Best in Class
            </span>
            <span className="text-xs font-semibold
text-[#5A6A72]">
                {metric.best_in_class} {metric.unit}
            </span>
        </div>
        <div className="h-2 bg-[#F4F6F7]
rounded-full overflow-hidden">
            <div className="h-full bg-[#1B3A4B]"
style={{ width: '100%' }} />
            </div>
        </div>
    </div>
</div>
);
}}
</div>
</CardContent>
</Card>

    {/ * Tabs for Different Visualizations */}
    <Tabs defaultValue="trends" className="w-full">
        <TabsList className="grid w-full grid-cols-3">
            <TabsTrigger value="trends">Historical
Trends</TabsTrigger>
            <TabsTrigger value="radar">Performance
Radar</TabsTrigger>
            <TabsTrigger value="peer">Peer Comparison</TabsTrigger>
        </TabsList>

        <TabsContent value="trends">
            <Card className="border-[#1B3A4B]">
                <CardHeader className="bg-gradient-to-r
from-[#1B3A4B] to-[#2E5A6B]">

```

```

        <CardTitle className="text-white">Performance
Trends</CardTitle>
    </CardHeader>
    <CardContent className="p-6">
        <ResponsiveContainer width="100%" height={400}>
            <RechartsLine data={historicalTrends}>
                <CartesianGrid strokeDasharray="3 3"
stroke="#F4F6F7" />
                <XAxis dataKey="period" stroke="#5A6A72" />
                <YAxis stroke="#5A6A72" />
                <Tooltip
                    contentStyle={{
                        backgroundColor: 'FFFFFF',
                        border: '1px solid #1B3A4B',
                        borderRadius: '8px'
                    }}
                />
                <Legend />
                <Line
                    type="monotone"
                    dataKey="current_venue"
                    name="Your Venue"
                    stroke="#9AAF45"
                    strokeWidth={3}
                    dot={{ fill: '#9AAF45', r: 6 }}
                />
                <Line
                    type="monotone"
                    dataKey="peer_average"
                    name="Peer Average"
                    stroke="#3A7B8A"
                    strokeWidth={2}
                    dot={{ fill: '#3A7B8A', r: 4 }}
                />
                <Line
                    type="monotone"

```



```

        dataKey="industry_average"
        name="Industry Average"
        stroke="#5A6A72"
        strokeWidth={2}
        strokeDasharray="5 5"
        dot={{ fill: '#5A6A72', r: 4 }}
      />
    </RechartsLine>
  </ResponsiveContainer>
  <div className="mt-4 p-4 bg-[#F4F6F7] rounded-lg">
    <p className="text-sm text-[#1B3A4B]">
      <strong>Key Insight:</strong> Your venue has
shown consistent improvement over
      the past year, moving from industry average to
top 10% performance in incident
      response time. The gap between your performance
and peer average has widened
      significantly in Q1 2025.
    </p>
  </div>
</CardContent>
</Card>
</TabsContent>

  <TabsContent value="radar">
    <Card className="border-[#1B3A4B]">
      <CardHeader className="bg-gradient-to-r
from-[#1B3A4B] to-[#2E5A6B]">
        <CardTitle className="text-white">Multi-Dimensional
Performance</CardTitle>
      </CardHeader>
      <CardContent className="p-6">
        <ResponsiveContainer width="100%" height={400}>
          <RadarChart data={radarData}>
            <PolarGrid stroke="#F4F6F7" />

```

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        <PolarAngleAxis dataKey="metric"
stroke="#5A6A72" />
        <PolarRadiusAxis stroke="#5A6A72" />
        <Radar
            name="Your Venue"
            dataKey="current"
            stroke="#9AAF45"
            fill="#9AAF45"
            fillOpacity={0.3}
            strokeWidth={2}
        />
        <Radar
            name="Peer Average"
            dataKey="peer"
            stroke="#3A7B8A"
            fill="#3A7B8A"
            fillOpacity={0.2}
            strokeWidth={2}
        />
        <Radar
            name="Industry Average"
            dataKey="industry"
            stroke="#5A6A72"
            fill="#5A6A72"
            fillOpacity={0.1}
            strokeWidth={2}
        />
        <Legend />
        <Tooltip />
    </RadarChart>
</ResponsiveContainer>
<div className="mt-4 grid grid-cols-3 gap-4">
    <div className="p-3 bg-[#9AAF45]/10 border
border-[#9AAF45] rounded-lg">
        <div className="text-xs text-[#5A6A72]
mb-1">Strengths</div>

```

```

        <ul className="text-xs text-[#1B3A4B]
space-y-1">
            <li>• Response Time (95th percentile)</li>
            <li>• Guest Satisfaction (94th
percentile)</li>
            <li>• Equipment Uptime (92nd percentile)</li>
        </ul>
    </div>
    <div className="p-3 bg-[#3A7B8A]/10 border
border-[#3A7B8A] rounded-lg">
        <div className="text-xs text-[#5A6A72]
mb-1">Competitive</div>
        <ul className="text-xs text-[#1B3A4B]
space-y-1">
            <li>• Prevention Rate (88th percentile)</li>
            <li>• Staff Efficiency (85th percentile)</li>
        </ul>
    </div>
    <div className="p-3 bg-orange-50 border
border-orange-500 rounded-lg">
        <div className="text-xs text-[#5A6A72]
mb-1">Opportunities</div>
        <ul className="text-xs text-[#1B3A4B]
space-y-1">
            <li>• Continue efficiency gains</li>
            <li>• Target best-in-class prevention</li>
        </ul>
    </div>
</div>
</CardContent>
</Card>
</TabsContent>

<TabsContent value="peer">
    <Card className="border-[#1B3A4B]">

```

```

        <CardHeader className="bg-gradient-to-r
from-[#1B3A4B] to-[#2E5A6B]">
        <CardTitle className="text-white">Peer Venue
Comparison</CardTitle>
    </CardHeader>
    <CardContent className="p-6">
        <ResponsiveContainer width="100%" height={400}>
            <BarChart data={peerComparisons}>
                <CartesianGrid strokeDasharray="3 3"
stroke="#F4F6F7" />
                <XAxis dataKey="venue_name" stroke="#5A6A72"
angle={-45} textAnchor="end" height={100} />
                <YAxis stroke="#5A6A72" />
                <Tooltip
                    contentStyle={{
                        backgroundColor: 'FFFFFF',
                        border: '1px solid #1B3A4B',
                        borderRadius: '8px'
                    }}
                />
                <Legend />
                <Bar dataKey="metrics.prevention_rate"
name="Prevention Rate (%)" fill="#9AAF45" />
                <Bar dataKey="metrics.satisfaction"
name="Satisfaction (x10)" fill="#3A7B8A" />
                <Bar dataKey="metrics.efficiency"
name="Efficiency (tasks/hr)" fill="#1B3A4B" />
            </BarChart>
        </ResponsiveContainer>

        {/* Peer Details Table */}
        <div className="mt-6 overflow-x-auto">
            <table className="w-full text-sm">
                <thead>
                    <tr className="border-b-2 border-[#1B3A4B]">

```

```

        <th className="text-left py-3 px-4
text-[#1B3A4B] font-semibold">Venue</th>
        <th className="text-left py-3 px-4
text-[#1B3A4B] font-semibold">Type</th>
        <th className="text-right py-3 px-4
text-[#1B3A4B] font-semibold">Capacity</th>
        <th className="text-right py-3 px-4
text-[#1B3A4B] font-semibold">Response Time</th>
        <th className="text-right py-3 px-4
text-[#1B3A4B] font-semibold">Prevention</th>
        <th className="text-right py-3 px-4
text-[#1B3A4B] font-semibold">Satisfaction</th>
    </tr>
</thead>
<tbody>
    {peerComparisons.map((peer, index) => (
        <tr
            key={peer.venue_name}
            className={`
                border-b border-[#F4F6F7]
                ${peer.venue_name === 'Our Venue' ?
'bg-[#9AAF45]/10 font-semibold' : ''}
                ${peer.venue_name === 'Best in Class' ?
'bg-[#1B3A4B]/5' : ''}
            `}
            >
                <td className="py-3 px-4 text-[#1B3A4B]">
                    {peer.venue_name}
                    {peer.venue_name === 'Our Venue' && (
                        <Badge className="ml-2
bg-[#9AAF45]">You</Badge>
                    )}
                    {peer.venue_name === 'Best in Class' &&
(
                        <Trophy className="inline h-4 w-4
ml-2 text-[#9AAF45]" />

```

```

        })
      </td>
      <td className="py-3 px-4
text-[#5A6A72]">{peer.venue_type}</td>
      <td className="py-3 px-4 text-right
text-[#5A6A72]">
        {peer.capacity.toLocaleString()}
      </td>
      <td className="py-3 px-4 text-right
text-[#1B3A4B]">
        {peer.metrics.response_time} min
      </td>
      <td className="py-3 px-4 text-right
text-[#1B3A4B]">
        {peer.metrics.prevention_rate}%
      </td>
      <td className="py-3 px-4 text-right
text-[#1B3A4B]">
        {peer.metrics.satisfaction}/5
      </td>
    </tr>
  </tbody>
</table>
</div>
</CardContent>
</Card>
</TabsContent>
</Tabs>

{ /* Improvement Recommendations */ }
<Card className="border-[#1B3A4B]">
  <CardHeader className="bg-gradient-to-r from-[#1B3A4B]
to-[#2E5A6B]">
    <CardTitle className="text-white">Path to Best in
Class</CardTitle>

```

```

</CardHeader>
<CardContent className="p-6">
  <div className="space-y-4">
    <div className="flex items-start gap-3 p-4
bg-[#9AAF45]/10 border border-[#9AAF45] rounded-lg">
      <Trophy className="h-5 w-5 text-[#9AAF45]
flex-shrink-0 mt-0.5" />
      <div>
        <h4 className="font-semibold text-[#1B3A4B]
mb-1">
          Target: Prevention Rate 72% (Currently 67%)
        </h4>
        <p className="text-sm text-[#5A6A72] mb-2">
          You're 5 percentage points away from
best-in-class performance. Based on peer
          analysis, focus on:
        </p>
        <ul className="text-sm text-[#1B3A4B] space-y-1
ml-4">
          <li>• Enhance predictive maintenance coverage
to 100% of critical equipment</li>
          <li>• Implement advanced crowd flow prediction
in high-density zones</li>
          <li>• Expand AI threat detection to include
behavioral analytics</li>
        </ul>
      </div>
    </div>

    <div className="flex items-start gap-3 p-4
bg-[#3A7B8A]/10 border border-[#3A7B8A] rounded-lg">
      <Target className="h-5 w-5 text-[#3A7B8A]
flex-shrink-0 mt-0.5" />
      <div>
        <h4 className="font-semibold text-[#1B3A4B]
mb-1">

```

```
        Competitive Advantage: Response Time
    </h4>
    <p className="text-sm text-[#5A6A72]">
        Your 4.2-minute average response time is 3x
better than peers. Document and share
        this capability with stakeholders to
demonstrate operational excellence.
    </p>
    </div>
</div>
</CardContent>
</Card>
</div>
);
};
```